

























































































































Today 11:54 PM



Lab #4\_ Beginning Blade ...  
PDF



Lab #6\_ Blade Bevel Lab....  
PDF



Tianwei Xi - Bladesmithin...  
PDF

```
<!DOCTYPE html>  
<html lang="en">
```

```
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width,  
initial-scale=1.0">
```

```
<title>Folding Knife Project | Tianwei Xi</title>
```

```
<link rel="stylesheet" href="../style.css">
```

Show more ▾

Yep. Based specifically on the three reports you uploaded, I'd revise the page quite a bit so it reflects what you can actually document.

A few important corrections: your project used **1084 steel**, the final CAD concept was a **drop-point folding design modeled in Fusion 360**, and the research specified a **liner-lock mechanism**.

Tianwei Xi - Bladesmithing Proj...

Your fabrication memo documents rough cutting, filing, blade shaping, and attention to heat buildup during sanding.

Lab #4\_ Beginning Blade Fabrica...

The later bevel lab documents comparing multiple bevel

geometries and ultimately selecting and producing a **sabre grind** using a jig for controlled, even sanding.

Lab #6\_ Blade Bevel Lab

I would **not claim that you personally completed quenching and tempering on this page yet**, because these three documents discuss heat treatment in the research portion but don't document your team actually completing that stage. I also changed SOLIDWORKS → **Fusion 360** because that's what your submitted project documentation shows.

Here is the replacement `folding-knife.html` :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Folding Knife Project | Tianwei Xi</title>

  <link rel="stylesheet" href="../style.css">
</head>

<body>

  <nav>
    <div class="nav-container">

      <a href="../index.html" class="logo">
```

This version is also stronger for an engineering portfolio because it frames the project around **materials selection → design development → CAD → manufacturing → engineering considerations**, rather than making the page read primarily as “I made a knife.”

One especially useful detail from your reports is that the **page 2 image in the bevel memo actually shows the jig being used with the belt sander and the completed bevel**.

Lab #6\_ Blade Bevel Lab If you have the original photos, I'd put 2–3 of them on this page:

**Fusion 360 model → rough-cut component → bevel/jig → finished project**. That would make the project much more convincing visually.

















Copy of Epic Quest 1 Tem...  
PDF

Yep — this report gives us much better metrics. I'd use numbers that show the **scale and technical scope** of the project rather than random individual measurements.

The report says you modeled a **2016 Subaru Forester**, collected position, velocity, acceleration, fuel-use, and related vehicle data, ran **3 trials for each of three courses**, and used the experimental data to develop MATLAB performance models.

Copy of Epic Quest 1 Template a... +1

For the **four metric boxes**, I recommend:

```
<div class="project-stats">

  <div class="project-stat">
    <strong>9</strong>
    <span>Course trials</span>
  </div>
```

```

<div class="project-stat">
  <strong>4</strong>
  <span>Performance models</span>
</div>

<div class="project-stat">
  <strong>0-35 mph</strong>
  <span>Baseline acceleration data</span>
</div>

<div class="project-stat">

```

## Why these four

**9 course trials** — Courses 1, 2, and 3 each had three experimental trials.

Copy of Epic Quest 1 Template a... +1

**4 performance models** — your report used the collected data to model the three experimental courses plus the Irwindale track.

Copy of Epic Quest 1 Template a...

**0–35 mph baseline** — the main performance dataset was collected by accelerating the Forester from rest to 35 mph and was then used as the basis for the predictive models.

Copy of Epic Quest 1 Template a...

**5 vehicle parameters** — position, velocity, acceleration, engine fuel rate/consumption, and RPM were the five main parameters considered. There were limitations with the RPM data, which your report explicitly discusses, so I'd say "**parameters analyzed**" rather than implying all five were successfully measured in every trial.

Copy of Epic Quest 1 Template a...

And there's actually enough substance in this report to make your **Vehicle Motion Data Analysis** portfolio page significantly better than the generic one we had before. You have OBD data collection, MATLAB modeling, kinematics, regression, three test courses, an Irwindale extrapolation, fuel-consumption modeling, and model-vs-experiment analysis.

Copy of Epic Quest 1 Template a...

If you want the same treatment as the knife project, I can populate your entire `vehicle-motion.html` from this report rather than just the metrics.

Sources







**Resume.pdf**  
PDF

Overall, it looks **strong and much more like an engineering resume now**. The one-page layout is clean, the hierarchy is easy to scan, and the projects now have actual technical substance and metrics. Looking at the rendered page, I would make a few targeted changes rather than redesigning it. Resume

## Biggest thing I'd fix: the knife project

Your current resume says **SOLIDWORKS**, but the project documentation you showed me earlier identifies the modeled blade as **Fusion 360**. Your current bullet also says you designed the liner-lock mechanism. Resume

I'd use:

### Folding Knife Fabrication Project

- Designed folding components and mechanical interfaces in Fusion 360, applying dimensional tolerances for proper fit, alignment, and assembly.
- Fabricated components from 1084 carbon steel through profiling, beveling, sanding, finishing, and final assembly.
- Performed quenching and tempering of 1084 steel to modify hardness and mechanical properties.

That's cleaner and avoids overstating anything.

## Vehicle project is one of your strongest sections

Keep this. The **9 trials + MATLAB + Irwindale extrapolation + numerical predictions** make it immediately more concrete than a generic school-project bullet. Resume

I'd only shorten the title to:

### Vehicle Performance Modeling & Data Analysis

That's more descriptive than "Vehicle Motion Data Analysis Project."

## Engineering Club section

The content is good, but it's taking up almost as much room as your technical projects.

Resume I'd tighten the wording slightly:

- **Social Media Manager:** Managed Instagram and Discord marketing for technical, professional, and social events, regularly attracting **50+ attendees**.
- **Inter-Club Council Representative:** Coordinated joint events with the Mathematics Club, Coding Club, Society of Women Engineers, and Astronomy Club.
- **Event Leadership:** Planned and executed a paper airplane engineering competition and game day, each attracting **50+ participants**.

Same accomplishments, fewer words.

## Education

I'd consider removing **high school** now that you're at UCR and have transferred from Mt. SAC. Currently, three schools take up a pretty large chunk of the top of the page.

Resume

Your SkillsUSA awards already preserve the strongest parts of your high-school accomplishments. Removing the high-school education line would give you space for another engineering project, another technical bullet, or simply more breathing room.

I'd eventually have:

**University of California, Riverside** — B.S. Mechanical Engineering | Sep 2026 – Present  
**Mt. San Antonio College** — Transfer | Jan 2022 – Jun 2026

And I'd probably change "Transfer" to something like **Engineering Coursework / Transfer Studies**, because "Transfer" isn't a degree.

## Skills

Good categories, but I would make the wording more engineering-specific. Currently you have CAD, fabrication, programming, and languages.

Resume

For example:

**CAD:** SOLIDWORKS, Fusion 360, AutoCAD, Onshape

**Engineering:** Mechanical Design, Materials Testing, 3D Printing, Laser Cutting, Power Tools

**Programming & Analysis:** MATLAB, Python, C++

**Languages:** English, Mandarin Chinese

I'd change "Programming" to **Programming & Analysis** because MATLAB is one of your strongest engineering-analysis tools.

## One thing I'd definitely keep

The **CSWP** belongs there. Resume Between CSWP, CAD software, your fabrication project, MATLAB vehicle modeling, internship, and UCR Mechanical Engineering, the resume now tells a pretty coherent engineering story.

The biggest improvement from here would be **freeing the high-school space and adding your Stirling Engine project**. That would give you three distinct technical projects: **experimental thermodynamics, computational/vehicle modeling, and design/manufacturing**, which shows considerably more range.

    ...  Sources

add the stirling engine project then

I'd add it under **Extracurriculars and Projects**, probably above Vehicle Performance Modeling because it's a strong experimental mechanical-engineering project.

### Stirling Engine Efficiency Analysis

- Conducted **10 experimental trials** of a Stirling-engine-powered cart over a **3.05 m track**, analyzing translational and rotational mechanical work.
- Combined experimental measurements with CAD-derived wheel properties to calculate **0.0433 J average mechanical work** and **0.0052% experimental efficiency**.
- Compared experimental performance with an **86.6% theoretical Carnot efficiency**, identifying heat-transfer losses, friction, and experimental assumptions as major sources of discrepancy.

For the resume, I'd probably use **only the first two bullets** if space is tight. They already give you excellent metrics: **10 trials, 3.05 m, 0.0433 J, 0.0052%**.

    ...  Sources

2 points

### Stirling Engine Efficiency Analysis

- Conducted **10 experimental trials over a 3.05 m track**, analyzing translational and rotational work to determine Stirling engine performance.
- Calculated **0.0433 J average mechanical work and 0.0052% experimental efficiency**, comparing results with an **86.6% theoretical efficiency** and evaluating major energy losses.

      Sources

ChatGPT is AI and can make mistakes. Check important info.